

1 Estimation of the Liquidity-Demand Shock

Following Telyukova (2013), we use a two-stage empirical procedure to estimate the volatility of innovations in consumer-unit liquid consumption. We then use this empirical moment, together with an adjacent-month consumption-correlation moment, to calibrate the idiosyncratic liquidity-demand shock θ_{it} in the model. The first-stage specification is

$$\log(c_{it}^{\text{liq}}) = \mathbf{X}_{it}'\beta + \mu_i + \varepsilon_{it}, \quad (1)$$

where $\mathbf{X}_{it}$ contains a cubic polynomial in age; indicators for education, marital status, race, and homeownership; annual earnings (wage and salary income plus self-employment income); family size; and calendar-month and calendar-year indicators. The term μ_i is a consumer-unit fixed effect, and ε_{it} is the idiosyncratic component of log liquid consumption. Although total income before taxes (FINCBTAX) is retained in the constructed FMLI data, it is not included in the maintained first-stage regression.

In the second stage, we estimate the empirical residual process

$$\varepsilon_{it} = \alpha\varepsilon_{i,t-1} + \eta_{it}, \quad (2)$$

using only consecutive-month residual pairs and no intercept. We use α exclusively for the persistence of the empirical first-stage residual, and η_{it} for its innovation. This notation distinguishes α from the model shock-persistence parameter ρ introduced below.

Let the model liquidity-demand shock follow

$$\log(\theta_{it}) = \rho \log(\theta_{i,t-1}) + \xi_{it}, \quad \text{Var}(\xi_{it}) = \sigma_\theta^2. \quad (3)$$

The estimated $\text{sd}(\hat{\eta}_{it})$ is interpreted as the empirical counterpart of the unconditional volatility of $\log(\theta_{it})$, rather than the volatility of the model innovation ξ_{it} . Writing $s_\eta = \text{sd}(\hat{\eta}_{it})$ and imposing covariance stationarity gives

$$s_\eta^2 = \text{Var}(\log(\theta_{it})) = \frac{\sigma_\theta^2}{1 - \rho^2}, \quad (4)$$

so that $\sigma_\theta = s_\eta \sqrt{1 - \rho^2}$. The model persistence ρ is calibrated separately by matching an adjacent-month liquid-consumption correlation, as described below.

The empirical measure c_{it}^{liq} is monthly consumer-unit expenditure in the maintained liquid-consumption basket, after category-specific CPI deflation and adjustment by DCPC liquid-payment shares. The benchmark basket contains food, alcohol, tobacco, rent, mortgage interest, repairs, insurance, and other owner-occupied dwelling expenses, utilities, household operations, childcare, property taxes, public transportation, and health insurance. Two sensitivity measures first exclude food, alcohol, and tobacco and then additionally exclude property taxes. Gift purchases

are excluded. Each category-specific monthly CPI is normalized by its 2019 mean before deflation.

Terminology. Throughout this document, a *Telyukova liquid-consumption group* (shortened after first use to *Telyukova group*) is the common aggregation level at which the CEX expenditure construction is matched to a DCPC liquid-payment share. A *CEX detailed category* and a *CEX component* are intermediate levels on the CEX side, while a *DCPC proxy category* is an observable merchant/follow-up classification on the DCPC side. These terms refer to distinct objects and are used consistently below.

1.1 Construction of Monthly Liquid Consumption c_{it}^{liq}

1.1.1 CEX Survey Catalog

We use the 2018 and 2019 CEX Interview Survey MTBI and FMLI files to construct monthly liquid consumption and estimate equations (1) and (2). In the parsed MTBI panel, each row identifies a consumer-unit interview, a reference month-year, and an expenditure category identified by a Universal Classification Code (UCC). The MTBI GIFT indicator identifies purchases for someone outside the consumer unit; we exclude these records as in Telyukova (2013). The FMLI files contain one record per consumer-unit interview and supply the demographic and income controls in $\mathbf{X}_{it}$.

The CEX crosswalk maps each selected UCC to a CEX detailed category, each detailed category to a CEX component, and each component to a Telyukova group. For each MTBI record, nominal expenditure is divided by the category-specific monthly CPI normalized to its 2019 mean and multiplied by the broad, day-of-week-weighted DCPC dollar share for the corresponding Telyukova group. The adjusted records are then summed within consumer unit and reference month to form c_{it}^{liq} . Months with no matched expenditure are set to zero, and only positive monthly values enter the log specification. The maintained construction and both regression stages are unweighted: the FMLI survey weight FINLWT21 is not applied.

Table 1 records the relationship among UCCs, CEX detailed categories, CEX components, and Telyukova groups for every component used in c_{it}^{liq} . Multiple UCCs appear in the same row only when they share the same downstream mapping.

Table 1: CEX UCC hierarchy used to construct Telyukova liquid-consumption groups

CEX UCC	CEX detailed category	CEX component	Telyukova group
190904, 790220, 790230, 190901, 190902, 190903, 790410, 790430, 800700	food_total	food	food_alcohol_tobacco_proxy
200900, 790310, 790320, 790420	alcohol	alcohol	food_alcohol_tobacco_proxy
630110, 630210	tobacco	tobacco	food_alcohol_tobacco_proxy
210110, 230121, 230141, 230150, 240111, 240121, 240211, 240221, 240311, 240321, 320611, 320621, 320631, 350110, 790690, 990920	rent_excluding_as_pay	rents	rent
800710	rent_as_pay	rents	rent
220311, 220313, 220321, 880110	mortgage_interest	mortgages	mortgage
210901, 220121, 220901, 230112, 230113, 230114, 230115, 230122, 230142, 230151, 230901, 240112, 240122, 240212, 240213, 240222, 240312, 240322, 320612, 320622, 320632, 340911, 990930	owned_dwelling_repairs_insurance_other	owned_dwelling_repairs_insurance_other	owned_dwelling_repairs_insurance_other
260211, 260212, 260213, 260214, 260111, 260112, 260113, 260114, 250111, 250112, 250113, 250114, 250211, 250212, 250213, 250214, 250221, 250222, 250223, 250224, 250901, 250902, 250903, 250904, 270101, 270102, 270103, 270104, 270211, 270212, 270213, 270214, 270411, 270412, 270413, 270414, 270901, 270902, 270903, 270904	utilities	utilities	utilities

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Table 1 – *continued from previous page*

CEX UCC	CEX detailed category	CEX component	Telyukova group
340310, 340410, 340420, 340520, 340530, 340903, 340906, 340910, 340914, 340915, 330511, 340510, 340620, 340630, 340901, 340907, 340908, 690113, 690114, 990900	household_operations_total	household_operations	household_operations
340211, 340212, 670310	childcare_within_household_operations	childcare	childcare
220211	property_taxes	property_taxes	property_taxes
530110, 530210, 530312, 530411, 530510, 530901, 530311, 530412, 530902	public_transportation	public_transportation	public_transportation
580111, 580112, 580113, 580114, 580311, 580312, 580901, 580903, 580904, 580905, 580906	health_insurance	health_insurance	health_insurance

1.2 Construction of Telyukova-Group Liquid-Payment Shares

The CEX records expenditure categories but not the payment instruments used for those expenditures. We therefore use the 2019 Diary of Consumer Payment Choice (DCPC) to estimate, for each Telyukova group, the share of expenditure paid with liquid instruments. Observable DCPC merchant and follow-up codes first define DCPC proxy categories; one or more proxy categories are then pooled into a Telyukova group. The resulting group-level share, rather than a proxy-category-specific share, is applied to the corresponding CEX components.

1.2.1 DCPC Survey Catalog

The 2019 DCPC is a three-day transaction diary covering consumer payments during October 2019. Each transaction-level record reports the respondent, diary day, amount, payment instrument, and merchant category. For selected merchant categories, additional pay codes identify the purpose or payee more precisely. The analysis retains payment transactions from diary days 1–3 that match an October 2019 diary-day record, have a positive amount and day-of-week weight, and have valid merchant and payment-instrument codes.

The maintained broad definition classifies cash, check, debit card, bank-account-number payment, online-banking bill payment, and account-to-account transfer as liquid payment instruments. Credit cards and all other payment methods remain in the denominator but not the liquid numerator. When a transaction uses multiple instruments, its dollar amount is allocated across the reported instruments; mixed transactions that cannot be completely parsed are excluded from the dollar-share calculation.

For Telyukova group g , let A_j be the amount of DCPC transaction j , let A_j^{liq} be the portion paid with broad liquid instruments, and let d_j be its day-of-week weight. The share applied in the CEX construction is

$$s_g = \frac{\sum_{j \in g} d_j A_j^{\text{liq}}}{\sum_{j \in g} d_j A_j}. \quad (5)$$

Thus, “weighted dollar share” below refers specifically to DCPC day-of-week weighting, not to the CEX survey weight `FINLWT21`. Each real CEX expenditure record assigned to group g is multiplied by s_g before consumer-unit-month aggregation.

Table 2 records the mapping from DCPC proxy categories to Telyukova groups. These are sometimes imperfect merchant-category proxies: for example, the DCPC full-mortgage-payment category supplies the share applied to the CEX mortgage-interest component, communications and entertainment supplement the utilities group, and taxi, air-travel, and delivery transactions supplement the public transportation group. The DCPC cash-contributions group is constructed for a separate sensitivity analysis but is not included in the maintained CEX benchmark basket.

Table 2: Mapping of DCPC proxy categories to Telyukova groups

DCPC proxy category	Telyukova group
food_grocery_proxy	food_alcohol_tobacco_proxy
food_restaurant_bar_proxy	food_alcohol_tobacco_proxy
food_fast_service_proxy	food_alcohol_tobacco_proxy
rent	rent
mortgage_payment	mortgage
utilities	utilities
communications_entertainment_proxy	utilities
household_operations_bill_proxy	household_operations
household_repairs	owned_dwelling_repairs_insurance _other
homeowners_renters_insurance	owned_dwelling_repairs_insurance _other
property_taxes	property_taxes
public_transport_tolls	public_transportation
taxi_air_delivery_proxy	public_transportation
health_insurance	health_insurance
childcare	childcare
cash_contributions	cash_contributions

The following flow diagram summarizes the two branches of the construction. They meet at the Telyukova-group level, where the group-level liquid-payment share is applied.

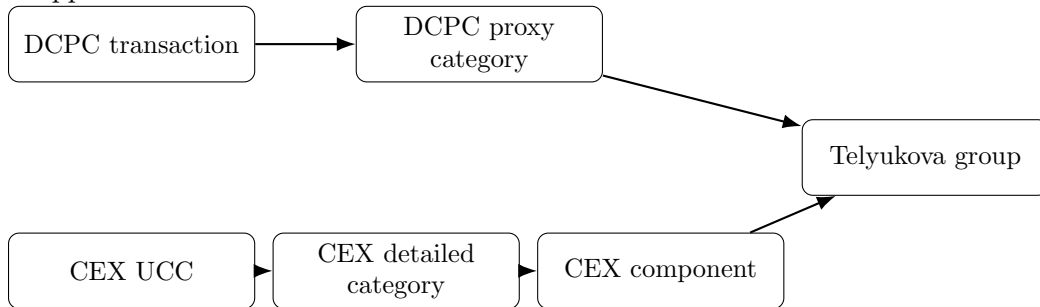


Table 3: Identification of DCPC proxy categories

DCPC proxy category	Merchant code and meaning	Follow-up identification rule and meaning
<code>food_grocery_proxy</code>	merch=1: Grocery stores, convenience stores without gas stations, and pharmacies.	None. The merchant code alone identifies the proxy category.
<code>food_restaurant_bar_proxy</code>	merch=3: Sit-down restaurants and bars.	None. The merchant code alone identifies the proxy category.
<code>food_fast_service_proxy</code>	merch=4: Fast-food restaurants, coffee shops, cafeterias, and food trucks.	None. The merchant code alone identifies the proxy category.
<code>rent</code>	merch=14: Rent for apartments, homes, or other buildings, including payments to real-estate companies and property managers.	None. The merchant code alone identifies the category.
<code>mortgage_payment</code>	merch=15: Financial-services providers, including mortgage companies, banks, credit-card companies, and insurance companies.	pay010=2: Purpose was to make a loan payment; and pay011=1: the loan was a mortgage.
<code>utilities</code>	Either merch=8: Nongovernment utilities, including electricity, natural gas, water, sewer, trash, and heating oil; or merch=19: Government taxes or fees.	For merch=8 , no follow-up is required. For merch=19: pay040=1 means a government purchase of goods or services, and pay041=1 identifies electricity, water, or sewer.
<code>communications_entertainment_proxy</code>	merch=10: Telephone, internet, cable or satellite television, streaming services, and movie theaters.	None. The merchant code alone identifies the proxy category.
<code>household_operations_bill_proxy</code>	merch=6: General services; or merch=11: Building contractors, plumbers, electricians, and HVAC providers.	from_bill_section=1: The transaction originated in the bill-reminder section. These records are assigned to household operations before the general contractor rule is evaluated.
<code>household_repairs</code>	merch=11: Building contractors, plumbers, electricians, and HVAC providers.	No additional follow-up. This category contains the remaining merch=11 transactions after the bill-reminder household-operations rule has been applied.

DCPC proxy category	Merchant code and meaning	Follow-up identification rule and meaning
homeowners_renters_insurance	merch=15: Financial-services provider.	pay010=3: Purpose was to pay for insurance; and either pay016=1: homeowners insurance, or pay016=2: renters insurance.
property_taxes	merch=19: Government taxes or fees.	pay040=2: Purpose was to pay taxes; and pay042=4: the tax was a property tax.
public_transport_tolls	Either merch=21: Public transportation and tolls; or merch=19: Government taxes or fees.	For merch=21 , no follow-up is required. For merch=19: pay040=1 means a government purchase of goods or services, and either pay041=5 identifies tolls or pay041=7 identifies public transportation.
taxi_air_delivery_proxy	merch=9: Taxis, airplanes, and delivery services.	None. The merchant code alone identifies the proxy category.
health_insurance	One of three merchant codes: merch=15 , financial services; merch=18 , medical-care providers; or merch=19 , government.	Three alternative rules: (1) merch=15 , pay010=3 for insurance, and pay016=3 for health insurance; (2) merch=18 and pay030=4 , identifying the payee as an insurance company; (3) merch=19 , pay040=1 for government goods or services, and pay041=8 for out-of-pocket health insurance, including Medicare supplemental insurance.
childcare	Either merch=20: Schools, colleges, and childcare centers; or merch=19: Government taxes or fees.	For merch=20 , pay020=3 identifies childcare. For merch=19 , pay040=1 identifies government goods or services, and either pay041=3 identifies daycare or pay041=9 identifies childcare.
cash_contributions	merch=17: Charitable or religious donations.	Either pay050=1: donation; or pay050=2: offering, tithe, or money placed in a collection plate.

1.3 Estimated Empirical Innovation Volatility

Table 4 reports $\hat{\alpha}$ and $\text{sd}(\hat{\eta}_{it})$ from equations (1) and (2) using monthly CEX data from 2018–2019. The first stage includes calendar-month and calendar-year indicators. Both stages and the reported innovation standard deviations are unweighted. The maintained output files use the legacy column name `rho` for $\hat{\alpha}$; the notation here distinguishes this empirical coefficient from the model persistence ρ .

Table 4: Monthly empirical innovation-volatility estimates, 2018–2019

Liquid-consumption measure	$\hat{\alpha}$	$\text{sd}(\hat{\eta}_{it})$	$\text{sd}(\hat{\eta}_{it})$ (%)
Benchmark	0.141	0.271	27.14
Excluding food, alcohol, and tobacco	0.139	0.285	28.53
Excluding food, alcohol, tobacco, and property taxes	0.109	0.325	32.47

1.4 Adjacent-Month Liquid-Consumption Correlation for Moment Matching

The model persistence ρ is calibrated by matching the correlation between liquid consumption in consecutive months. Table 5 reports the available 2019 candidate moments in levels and logs. These are pooled, unweighted correlations across strictly consecutive months for which both consumer-unit observations have positive liquid consumption.

Table 5: Adjacent-month liquid-consumption correlations, 2019

Liquid-consumption measure	Adjacent pairs	Consumer units	$\text{Corr}(c_{it}^{\text{liq}}, c_{i,t-1}^{\text{liq}})$	$\text{Corr}(\log c_{it}^{\text{liq}}, \log c_{i,t-1}^{\text{liq}})$
Benchmark	51,095	11,432	0.589	0.852
Excluding food, alcohol, and tobacco	50,951	11,397	0.560	0.844
Excluding food, alcohol, tobacco, and property taxes	50,913	11,387	0.509	0.810